

Greenhouse Effect, Atmospheric Solar Absorption and the Earth's Radiation Budget: From the Arrhenius-Langley Era to the 1990s

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Greenhouse Effect, Atmospheric Solar Absorption and the Earth's Radiation Budget: From the Arrhenius-Langley Era to the 1990s

In his 1896 paper, Svante Arrhenius laid the foundation for the modern theory of the greenhouse effect and climate change. The paper is required reading for anyone attempting to model the greenhouse effect of the atmosphere and to estimate the resulting temperature change. Arrhenius demonstrates how to build a radiation and an energy balance model directly from observations. Arrhenius was fortunate to have access to Langley's data, which are some of the best radiometric observations ever undertaken from the surface. The successes of Arrhenius' model are many, even when judged by modern-day data and computer simulations: the suggestion of the diffusivity factor including its correct numerical value; the remarkably accurate simulation of the total emissivity of the atmosphere which seem to agree within 5% of modern-day values; the logarithmic dependence of the CO₂ radiative heating effect; and others documented in the text. We uncover two aspects of the model which, most likely, were not recognized by earlier studies: First, Arrhenius included the water vapor feedback by introducing the fixed relative humidity assumption, which amplified the surface warming in his model by about 30%. Second, his model overestimated the surface warming, primarily because the radiation model overestimates the opacity of the CO₂ bands in the 6 to 8 μm region. In constructing his model, Arrhenius had to account for the magnitude of the solar radiation absorbed within the atmosphere—a topic that is currently pursued with renewed vigor. The second part of the paper addresses this topic, including the controversy that surrounds it. Observed values of solar absorption, since the 1950s, have almost always exceeded theoretical and model values. The magnitude of this excess absorption, i.e., observed-theoretical absorption, on climatologically relevant time and spatial scales was quantified recently (1990s) by six independent studies, to be about 25 W m⁻² or larger. We review some of the key results published between the 1950s to 1990s to clarify the central issues and offer suggestions for resolving the discrepancy between observations and models.

INTRODUCTION

The greenhouse effect of the atmosphere was known as early as 1824 (1), e.g., see Fourier (2). A formal foundation for the theory linking atmospheric gases with climate change had to await the publication of Professor Svante Arrhenius' paper in the year 1896. The main goal of this paper was to estimate the surface temperature increase due to an increase in CO₂. Towards this goal, Arrhenius developed a detailed and quantitative model for the radiation budget of the atmosphere and the surface. To a large extent, the model relied on surface radiometric observations, particularly those of Samuel Pierpont Langley. Langley (3, 4) invented a new high precision thermal detector, the bolometer, and recorded numerous observations of the lunar and solar spectra. Langley recorded both the broad band radiance and the spectral radiance in about 20 spectral bands between 0.9 to 30 μm .

Arrhenius deduced a number of fundamental parameters, including: i) wavelength dependent absorption of solar radiation (light heat) by water vapor and carbon dioxide; ii) Rayleigh scat-

tering (selective reflection); iii) absorption of terrestrial or IR radiation (dark heat) by H₂O and CO₂; iv) the so-called diffusivity factor (mean path of Earth's rays); v) mean cloud fraction (0.525); vi) the logarithmic dependence of the CO₂ greenhouse effect; and vii) albedos of sea surface, land, clouds, old and new snow. The next section reviews briefly Arrhenius' work on the greenhouse effect of water vapor and carbon dioxide. Our main objective in the next section is to demonstrate the important role played by the water vapor in Arrhenius' model, including the role of the water vapor feedback in the predicted surface temperature change due to CO₂ increase.

It is a remarkable coincidence that, during the 100th anniversary of the publication of this paper, the climate and the radiation budget communities are actively scrutinizing surface solar observations to explain a major discrepancy between climate models and radiometric observations. At dispute is the fraction of the solar radiation absorbed by the atmosphere—a topic that was considered in great detail by Langley and Arrhenius. In fact, the opening section of Arrhenius' paper is titled: "I. Introduction: Observations of Langley on Atmospheric Absorption." The main focus of the last two sections of our paper is on the atmospheric solar absorption.

The rest of this introductory section is devoted to the following question: How has the field advanced in the intervening 100 years?

First, the dawn of the computer era in the 1960s made it possible for us to develop detailed radiation models that assimilated spectroscopic and quantum mechanical parameters, e.g., see Goody (5), as opposed to radiances in Arrhenius' time. The ability of these models to generate radiation fluxes far outstripped our ability to make such observations (e.g., the vertical distribution of the fluxes), such that models, particularly those used in climate studies, were developed in isolation. Symptomatic of this process is the treatment of clouds as flat plates with no inhomogeneities.

The above picture changed dramatically with the advent of the weather satellite in the 1970s, particularly, since radiation budget observations were some of the first data collected by satellites, thanks to the pioneering efforts of Professor V. Suomi and his colleagues at the University of Wisconsin. It took us nearly two decades to resolve many of the obstacles in interpreting the satellite radiation budget data. Today, we are back to Arrhenius' era where one uses radiances to deduce radiation budget quantities and climate model parameters. The only difference from that era is that radiances were obtained above the Earth from satellites.

With a satisfactory picture of the planet from above, the trend now is to combine the view from above with the view from below (radiation data from the surface) and within the atmosphere through aircraft data. Recent attempts, by at least six different international groups (6–11; amongst several others) at compositing satellite data with surface and in situ data have unearthed a major discrepancy between observations and models in the estimated transparency of the atmosphere to solar radiation—a topic which is alternately known as "excess absorption" or "anomalous absorption", e.g., Stephens and Tsay (12). In this context, we will review the historical development of this topic including consideration of the following question: How does the

Table 1. Terms used by Arrhenius with the corresponding modern-day terms.

Arrhenius' Term	Corresponding Modern-day Term	Explanation
Selective absorption	Absorption bands	Absorption that occurs in discrete wavelengths
Absorption coefficient, X and Y in Table II	Transmission in a narrow spectral interval band. The corresponding absorption coefficient, k, as it is referred to now, can be obtained by letting: $k = -\ln(x)$ or $-\ln(y)$	X or Y is the fraction of incident radiation transmitted by unit mass of CO ₂ in the case of X and unit mass of H ₂ O in the case of Y
Transparency in Table III	Broad band transmittance; the emissivity is = 1. -transparency	Obtained by weighting the narrow band transmission, or the "absorption coefficient" in Arrhenius Table II with the black body Planck function within each band
Mean path of the Earth's rays in Table IV	Diffusivity factor	This diffusivity factor approximates the angular integration of the transmission
b, the 'absorption coefficient' for the heat that radiates from the surface; this term is used in the climate model, Eqs. 1 to 3	Emissivity (see the earlier entry for transparency). This variable is used to characterize the emission of absorption of infrared radiation by a surface or a layer of gas	The fraction of black body radiation emitted by a body or a layer of gas; for an isothermal atmosphere, it also represents the fraction of the incident black body radiation absorbed by the body or a layer of gas
n, albedo of Earth's crust	Surface albedo	Fraction of solar radiation reflected by the surface
Heat conducted by the atmospheric and oceanic currents, horizontal or vertical	The word "heat" in this context refers to sensible and latent heat	Sensible heat refers to the kinetic energy due to molecular motions; latent heat is the heat that will be released if all of the moisture in the parcel of air is condensed to water or ice

opacity of the atmosphere to light and dark rays inferred from modern-day observations compare with Arrhenius' and Langley's data?

ARRHENIUS' MODEL FOR THE ATMOSPHERIC GREENHOUSE EFFECT

Table 1 lists the terms used by Arrhenius with the corresponding modern-day terms.

Assumptions of Arrhenius' Model

It should be recognized that Arrhenius did not use the model to compute the temperature of the planet. Many of the assumptions in his model are not appropriate for this purpose. He used his model primarily for estimating the temperature change due to a change in the concentrations of H₂O and CO₂, and his model assumptions, listed below, are not so limiting for this application.

I. Fixed dynamical heat transport. Horizontal transport of heat by "currents" in the air and sea do not change with a change in temperature; this assumption enabled Arrhenius to compute the surface temperature change as a function of latitude. In modern-day terminology, the term "heat" as it is used in Arrhenius' paper, includes sensible heat (kinetic energy of molecular motions) and latent heat (the heat that will be released if all of the moisture in the air is condensed to water or frozen to ice).

II. Fixed vertical heat exchange between air and sea or land. This assumption, which implicitly accounts for the sum of the latent and sensible heat fluxes from the surface, enabled him to compute the surface temperature change without getting into complex calculations of the latent heat fluxes.

III. Fixed relative humidity. The greenhouse effect of water vapor increases with an increase in surface temperature, i.e., Arrhenius allowed for the water vapor feedback in a manner similar to the modern-day radiative-convective models. We will return to this topic later.

IV. Fixed clouds and surface albedo. There was no cloud feedback or ice/snow albedo feedback, although he recognized the importance of the ice-albedo feedback. Even the modern-day one-dimensional radiative-convective models do not account for such feedbacks.

The Model Type

The model has two equations for two unknowns: The surface temperature and the atmospheric temperature. Thus, it allows for the coupling between the surface and the atmosphere. It is essentially a one-dimensional model. There are two types of one-dimensional models: a) the radiative equilibrium model (13), in

which the nonradiative fluxes, including the vertical and the horizontal transport of sensible and latent heat fluxes are set to zero; and b) the radiative convective equilibrium model (14), in which the net radiative flux is balanced by the so-called convective heat flux (vertical transport of sensible and latent heat fluxes). Because of assumptions I and II above, Arrhenius' model does not conveniently fall into either of the two categories. It is a hybrid of the two models.

– By allowing for an explicit heat transport term (the M term in his Eq. 1 and the N term in his Eq. 2), he essentially anticipated the radiative-convective models pioneered by Manabe and his coworkers.

– However, since the heat transport terms M and N are held fixed while increasing CO₂, the departure from equilibrium is assumed to follow radiative equilibrium. This type of radiative equilibrium adjustment model with fixed dynamical heating rates, is now routinely used (e.g. 15, 16).

Validity of the Model Parameters

We will focus on those parameters that are critical to the surface temperature calculations.

H₂O and CO₂ Absorption Coefficients

The model includes almost all of the important absorption bands ("selective absorption") from 0.9 μm to about 30 μm (see Table II of Arrhenius). The following three points should be kept in mind when relating Arrhenius' discussions in terms of modern-day terminology:

– The gaseous absorption, which occurs usually in discrete wave lengths, was referred to then (1900s) as "selective absorption" and is now referred to as absorption bands. The absorption bands in the visible and the longer wave lengths occur as a result of vibrational and rotational motions of molecules (around the center of axis) and are referred as vibration-rotation bands. When only rotational motions are involved, the bands are referred to as "pure rotation bands".

– Table II of Arrhenius, following Langley, lists the absorption coefficients as a function of the "angle of deviation" of a rock salt prism. This is the angle of deviation of the incident radiation beam produced by the prism (used in the measurement) due to refraction. The angle of deviation increases with decreasing wavelength, and hence is a measure of the wavelength. In the text following Table II, Arrhenius converts the angle into wavelength.

– Arrhenius lists values for log₁₀X, where X is referred as absorption coefficient. In modern-day use, X is the transmission or "transparency" (see Arrhenius' Table III). The absorption

coefficient, k can be obtained from the X given in Arrhenius' Table II by letting: $k = -\ln X$, where $\ln$ is the natural logarithm ($\log_e$).

With respect to water vapor, Arrhenius' model includes the vibration-rotation bands centered in the 0.94 μm , 1.4 μm , 1.8 μm , 2.7 μm , 6.3 μm , and the pure rotation band beyond 25 μm region. For CO_2 , it includes the vibration-rotation bands at 1.4 μm , 1.6 μm , 2 μm , the strong 2.7 μm , the super strong 4.3 μm , 5 μm , 10 μm , and the strong 15 μm bands.

Between the 5 to 10 μm , the CO_2 absorption by the Arrhenius' model is too strong. In addition, the 15 μm band absorption extends almost up to the 20 μm region with the transmission at 20 μm almost as strong as in the 15 μm region. This is undoubtedly due to the fact that in these spectral regions, H_2O and CO_2 bands overlap and it is difficult to separate the two effects solely from surface observation. It is our guess that some of the H_2O absorption effects are attributed to CO_2 . There is at least a factor of 2 overestimation of the absorption in this region, and this is perhaps one of the main reasons for the overestimation of the CO_2 greenhouse effect in Arrhenius' model. Similarly the water vapor absorption in the continuum region between 8 to 12 μm region has been overestimated significantly.

Figure 1 compares the combined H_2O plus CO_2 transmittance from Arrhenius' model with one of the recent longwave atmospheric models (17). Arrhenius gives the H_2O plus CO_2 transparency (transmittance). In his energy balance model, however, he refers to this term, $\{1. - \text{Transparency}\}$, as "absorption coefficient". The "absorption coefficient", b , in his climate model is the same as "emissivity" (see Table 1).

– We had to make the following assumptions to convert Arrhenius' Table III to the numbers given in Figure 1. Arrhenius uses surface humidity as a measure of column water vapor, and his symbol is W for water vapor column amount. It is defined such that when $W = 1$, the surface humidity is 10 g m^{-3} . In order to convert W to column water-vapor, we need to know the water-vapor scale height. We assumed a mean scale height of 1.7 km (it can vary from 1.5 to 2.2 km), such that, $W = 1$ denotes a column water vapor amount of 17 kg m^{-2} . For CO_2 , Arrhenius' Table III, uses a normalized value of 1 to denote the observed column CO_2 amount, presumably, for the years 1884–1887 (the years of Langley's observation). For the comparison in Figure 1, we used the transmission values for CO_2 values of 1.25 (in Arrhenius' Table III) since our model uses a CO_2 amount of 340 ppm, which is 1.25 times the pre-industrial value of 270 ppm.

– Our model uses over 4000 vertical radiosonde soundings of water vapor over mid-latitude regions to compute the transmission and the emissivity of the atmosphere from the surface to the top-of-the atmosphere. This simulates Langley's observations of the total transmission of the atmosphere between the surface

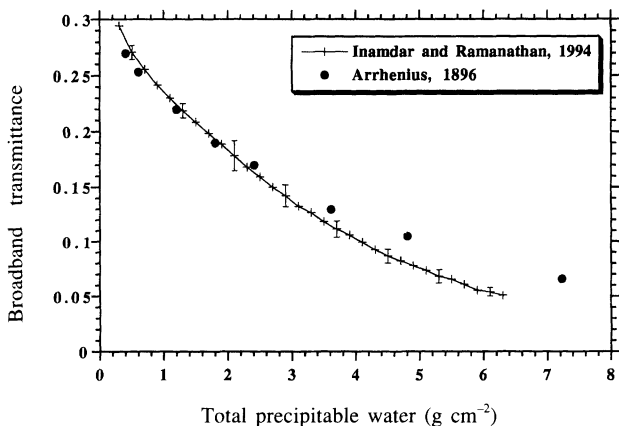


Figure 1. Comparison of Arrhenius' transmittance with a recent model calculation.

and the sun or the moon. Furthermore, the model computes the diffuse transmission. As pointed out by Arrhenius, the diffuse transmission can be obtained by multiplying the water vapor amounts by 1.66. This effect was included as follows. In order to obtain Arrhenius' water vapor transmission for $W = 1$, we adopted the transmission for $W = 1.66$ (in his Table III). The same procedure was followed for CO_2 .

The agreement between the modern-day models and Arrhenius' empirical transmission values is remarkable. The only deficiency is in apportioning the total absorption (or transmission) between H_2O and CO_2 .

Diffusivity Factor or the "Mean Path of the Earth's Rays"

The first step in the greenhouse effect calculations involves the estimation of the tedious zenith angle integral of the transmission function, given by:

$$T^f(\tau) = 2 \int_0^{\frac{\pi}{2}} e^{-\frac{\tau}{\mu}} \cdot \mu \cdot d\mu \tag{Eq. 1}$$

where $e^{-\frac{\tau}{\mu}}$ is the transmission at a zenith angle μ for an optical depth of τ and $T^f(\tau)$ is the diffuse transmission function. It is now conventional (18) to approximate Eq. 1 by:

$$T^f(\tau) = 2 \int_0^{\frac{\pi}{2}} e^{-\frac{\tau}{\mu}} \cdot \mu \cdot d\mu \approx e^{-1.66\tau} \tag{Eq. 2}$$

where 1.66 is the so-called diffusivity factor or as referred to more accurately by Arrhenius as "the mean path of the Earth's rays". This approximation has been shown to be accurate for water vapor and CO_2 bands. Arrhenius seems to be the first to have suggested this approximation. Furthermore, he correctly concluded (see his Table IV) that the diffusivity factor should increase from about 1.4 at large values of t to 2 for the optically thin limit. As a mean value for current-day values of water vapor and CO_2 , he used a mean value of about 1.61.

Other Radiation Parameters

He adopted a mean value of 0.525 for cloud amount, 0.55 for cloud albedo, 0.78 for fresh snow albedo, 0.5 for old snow albedo, 0.075 for ocean albedo, surface relative humidity of about 75 to 80%. All of these values, or ones close to them, are still acceptable and are used in many models.

Assessment of the Predicted Surface Temperature Change

Arrhenius' solution for the surface temperature is given by (His Eq. 3):

$$T^4 = \frac{k}{\{1. + v(1. - \beta)\}} \tag{Eq. 3}$$

where T is the surface temperature, and k is a sum of several terms including solar absorption, all of which are assumed to be independent of T . In the denominator, v is co-albedo of the surface, i.e., $1. -$ surface albedo, and β is the atmospheric emissivity. Arrhenius refers to β as the absorption coefficient, but it is equivalent to the emissivity because he obtains β , by weighting the fractional absorption ($1. -$ transmission) in each sub-band with the black-body energy within each band. Since the spectral Planck function was not yet discovered, Arrhenius obtained the energy within each band from Langley's spectral bolometric measurements.

As pointed out by Arrhenius, Eq. 3 shows that T increases with $\beta \dots$ a clear demonstration of the greenhouse effect. Arrhenius computed the surface temperature change numerically from Eq. 1. In this regard, we emphasize an often ignored point. Arrhenius accounted for the water-vapor greenhouse effect, and we quote from Section IV, p. 263 of Arrhenius to prove our point. In the

following quotes, K and W are, respectively, the CO₂ and H₂O concentrations for the unperturbed atmosphere, and ρ is the diffusivity factor which varies slightly with the concentrations. The subscript 1 refers to the atmosphere with increased CO₂.

“Suppose now that the carbonic acid had another concentration K₁ (e.g. K₁ = 1.5). Then we first suppose W unaltered. . . . Next we have to seek β, which corresponds to ρ₁ K₁ (1.5 ρ₁) and ρ₁ w.” Step 1

Arrhenius solved Eq. 3 numerically, one step at a time. Hence, we denote the above as Step 1. First he solves for the T, with the new value of CO₂ but holding the W fixed. The next sentence from Arrhenius, clearly shows how he accounted for the water vapor feedback: he computes a new W for the new temperature, by holding the relative humidity fixed.

“In consequence of the variation (t) in the temperature, W must also undergo a variation. As the relative humidity does not vary much, . . . , we have supposed that this quantity remains constant, and thereby determined the new values W₁ of W.” Step 2.

We now proceed to evaluate the role of the water vapor feedback in Arrhenius’ model. We can solve Eq. 3 analytically, by taking the natural log of both sides of Eq. 3 and differentiating it with respect to K (the CO₂ concentration) after noting that the emissivity β varies with K and W.

$$\Delta T_{\text{Step1}}(2 \text{ CO}_2) = \frac{\alpha T}{4} \Delta \beta(2 \text{ CO}_2); \text{ without H}_2\text{O feedback} \quad \text{Eq. 4a}$$

$$\Delta T(2 \text{ CO}_2) = \frac{\Delta T_{\text{Step1}}(2 \text{ CO}_2)}{[1 - (\frac{\alpha T}{4}) \{ (\frac{\partial \beta}{\partial \ln w}) (\frac{d \ln w}{dT}) \}]}; \quad \text{Eq. 4b}$$

with H₂O feedback

where $\alpha = \frac{v\beta}{1+v(1-\beta)} = 0.627; \quad \text{with } v = 0.925; \beta = 0.8$

Equation 4a ignores the water vapor feedback, i.e., dW/dT = 0.0., and it constitutes Step 1 of Arrhenius’ solution technique. From his Table III, we obtain: Δβ(2CO₂) = β(3.2 CO₂) – β(1.6 CO₂) = 0.084. This value of Δβ is too large by about a factor of 2.5, compared to modern calculations, presumably due to the overestimation of the absorption in the 5 to 7 μm and 18 to 20 μm regions (as pointed out earlier). Upon substituting this value in Eq. 4a, we obtain:

$$\Delta T_{\text{Step1}}(2 \text{ CO}_2) = 3.8 \text{ K}; \quad \text{without H}_2\text{O feedback} \quad \text{Eq. 5a}$$

The water feedback terms in the denominator of Eq. 4b are: {dlnW/dT} ≈ 0.06 K⁻¹ for fixed relative humidity and {dβ/dlnW} = 0.097 from Table III of Arrhenius and substituting in Eq. 4b, we obtain :

$$\Delta T_{\text{Step2}}(2 \text{ CO}_2) = 5.13 \text{ K}; \quad \text{with H}_2\text{O feedback} \quad \text{Eq. 5b}$$

The estimate given by Eq. 5b agrees almost exactly with Arrhenius’ results, confirming the accuracy of our analytical derivations. Upon comparing Eq. 5a with 5b, we conclude that water feedback amplified the surface warming by a factor of about 1.3 in Arrhenius’ model. If we adjust for the factor of 2.5 overestimation of Δβ due to CO₂ increase, the warming estimates from Arrhenius model would be 1.5 K without H₂O feedback and 2 K with H₂O feedback . . . values almost identical to those yielded by modern-day one dimensional models.

ATMOSPHERIC SOLAR ABSORPTION

How much solar radiation is absorbed in the clear and cloudy atmosphere? The answer to this question is not only of intrinsic scientific importance, but also is needed to estimate the energy

absorbed by the surface of the Earth. We review papers published since the 1960s to document the fact that observed values of atmospheric solar absorption almost always exceeded theoretical or model values. This systematic positive difference between the observed atmospheric solar absorption and the theoretical/model estimates is referred to here as “excess absorption.” Until the early 1980s, it was generally agreed that there was indeed some anomalous absorption features in the atmosphere which cannot be accounted for by standard theory or models (e.g., see the reviews of earlier work in refs. 12 and 18). Furthermore, some of the studies were indicating that a fraction of this absorption may be happening at visible wavelengths (e.g., 19–21) where the atmospheric absorption is weak and clouds were supposed to be conservative scatterers.

During the 1980s, it was generally thought that the absorption feature in the visible region was an artifact of the aircraft data introduced by imperfect sampling of the three-dimensional radiation field of clouds (22). An ad hoc normalization procedure was developed to subtract out the fractional excess visible absorption from both the visible and the near-IR (e.g., 22, 23). This procedure essentially removed all of the excess absorption and brought the observations closer to the modeled values, instead of the other way around, leading to the currently prevailing view that there is no major discrepancy between models and observations and that the excess absorption, if it exists, is only marginal .

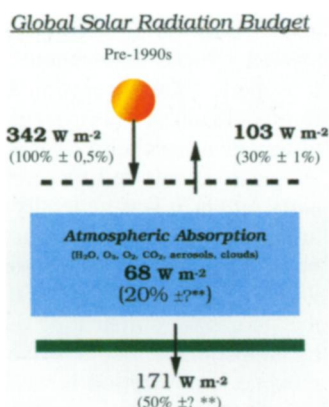
However, the advent of accurate radiation budget satellites with onboard calibration, and the ability to combine them with accurate surface data, made it possible to estimate the atmospheric absorption directly without resorting to aircraft data. Several such studies (6–8, 11, 21) not only supported the existence of excess absorption, but also quantified its magnitude on climatologically relevant time and space scales to be about 25 to 30 W m⁻² (diurnal, annual average), which represents an additional 8% absorption of the top-of-atmosphere insolation. A stacked aircraft study conducted simultaneously arrived at the same conclusion (9). In addition, the Cess et al. (7), Ramanathan et al. (8), and Pilewskie and Valero (9) studies (hereafter referred as C95; R95 and PV95 respectively; the “95” denotes the year of the publication of these papers) concluded that the cloudy sky (as opposed to clear sky) was the source for the excess absorption.

The large magnitude, the lack of an adequate physical explanation for the excess absorption, and conflicting results from the various studies have become the source of a major debate (11, 24–27). The main goal of this section is to provide a background for understanding the nature of the discrepancy between models and observations.

Discrepancy Between Observations and Models: Global Magnitude

Satellite radiation budget measurements over the last 15 years have confirmed that the top-of-the atmosphere (TOA) insolation is 342 W m⁻² and the TOA albedo is 30% (28–31). Figure 2 shows these observed values (with uncertainties) along with the numbers quoted in various sources including text books (32, 33) for the atmospheric absorption (20%) and surface absorption (50%). The numbers shown in Figure 2 are for average cloudy sky conditions (a global climatological average of cloud cover and type). The precise data source for the surface and atmospheric absorption is somewhat obscure, but it is safe to assume that it is based on a mixture of observations, modeling studies, extrapolations, and unspecified assumptions. For example, radiometric observations over the oceans are rare. However, radiation model calculations, which account for the three dimensional distribution of atmospheric gases as well as the latitudinal and the seasonal variation of the insolation, support the value of 20% for atmospheric absorption (34).

Figure 2. Current estimate of the global annual mean solar radiation budget for average cloud cover. All of the uncertainties in this figure and other figures are 2σ values. The uncertainties in TOA insolation and TOA albedos are based on the various published estimates of the two quantities by the earth radiation budget satellite studies during the 1970s to 1990s (28–30). **The uncertainties in atmospheric and surface absorption values are not specified in the cited references. (Source: 32, 33).



The results of the recent observational studies (6, 11, 21, C95, R95, PV95) suggest a different view as summarized in Figure 3. These studies suggest that the atmosphere absorbs as much as 28%, which is balanced by a corresponding reduction in the surface absorption to about 42% instead of the 50% value shown in Figure 2.

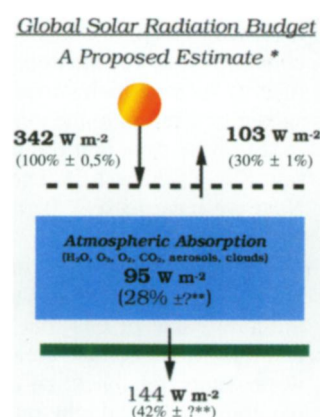
We begin with the data from the global surface observational network, the Global Energy Budget Archive (GEBA) data, analyzed by the data center at Zurich by Ohmura and Gilgen (6). While this network is extensive over the continents, the data coverage over the ocean is minimal. According to this data set, only 42% of the TOA insolation is absorbed at the surface. In terms of energy units, the solar radiation absorbed at the surface is about 25 W m⁻² less than the previous estimate shown in Figure 2. In turn, the atmosphere absorbs about 28%, i.e., 8% more than the previous estimates. The new data were compared to general circulation models (GCMs) in an important collaborative study between the experimental group at Zurich and the GCM modeling group at Hamburg (10). This study, which compared output from nine GCMs with the new data set revealed that these GCMs overestimate the surface solar absorption by about 20 to 30 W m⁻² (Fig. 4).

The other studies that use different combinations of aircraft, satellite, and surface measurements came to the same conclusion (C95; R95; PV95 and 11). A comparison study by Gleckler and Weare (35), which included 30 GCM simulations for the world oceans, concluded: "Preliminary comparison of zonal average fluxes suggest that most atmospheric general circulation models produce excessively large ocean surface fluxes of net solar heating..."; in addition it pointed out that "... in the mid-latitudes very few models have shortwave fluxes in the range of the observations."

When was this discrepancy discovered? A systematic discrepancy between models and observations has existed for nearly four decades. For example, Robinson (19) and Roach (36) showed that the atmospheric solar absorption, inferred from surface and atmospheric radiometric observations, is systematically larger than that estimated by models. Herman (37) and Herman and Curry (20) studied aircraft data of Arctic stratus clouds and found absorption at visible and near-IR wavelengths that significantly exceeded computations. Twomey and Cocks (38) also used aircraft data to show that the near-IR absorption in clouds is significantly larger than that predicted by models. Most of these earlier studies are summarized in Stephens and Tsay (12) and Liou (18). This persistent and systematic discrepancy between theory and observations was named "Anomalous Absorption" (12), because the source for the larger absorption was not known—and is still unknown. However, for reasons discussed later, sufficient doubt persisted in the data such that the discrepancy was left unresolved.

Does the discrepancy exist in cloudy or clear skies? There are diverging results on this issue. C95, R95 and PV95 concluded that most, if not all, of the discrepancy is in the cloudy skies. Furthermore, these studies show that the solar absorption in

Figure 3. Proposed estimate of the global-annual mean solar radiation budget for average cloud cover. **The proposed values are based on Ohmura and Gilgen (6), and Cess et al. (7), which are supported by Ramanathan et al. (8), Pilewskie and Valero (9). The global magnitude of the uncertainties are not estimated in the cited references.



cloudy skies is significantly larger than that in clear skies, whereas in models, the cloudy sky solar absorption is nearly the same as that in clear skies. However Arking (11), among others, concluded that all of the 25 to 30 W m⁻² discrepancy in atmospheric absorption is in clear skies. The debate surrounding the most recent studies, in our opinion, appears to stem from several factors:

- The magnitude of the cloudy sky absorption in the C95, R95 and PV95 studies is too large compared to expectations from the models, and perhaps also from prior observational estimates of excess absorption (24).

- Many aircraft studies (22, 23, 39) do not support the findings of large excess absorption. We note here that observations from aircraft observations are the most direct way of observing cloud solar absorption.

- Most of the studies prior to the C95, R95 and PV95, concentrated mainly on a single cloud or a single cloud system (12). While these studies also found systematic discrepancies, these were in terms of cloud properties (inferred single scattering albedos, absorption coefficients, etc.; 12) of a single cloud. For example, Liou (18) summarizes the earlier studies as follows:

"Reflectance and absorptance (of clouds) computed from theoretical programs . . . are generally higher and lower respectively than observed data. The largest cloud absorptance derived from theoretical calculations has an upper limit of $\approx 20\%$. . . However, the observed cloud absorptance could be larger than 30%."

Thus, the overall magnitude of the problem on climatologically relevant time and space scales was not obvious from the studies prior to C95, R95 and PV95.

- Since the earlier studies looked at isolated cloud systems, the discrepancy was attributed to (with few exceptions such as the Twomey and Cocks (38) study) instrumental errors (12), or radiation leaking through the cloud sides that are difficult to measure from aircraft (22, 39). The sampling was questioned since the experiments typically used a single aircraft and the cloud absorption, derived from differences between the flight legs above and below cloud, could just be an artifact of the cloud internal inhomogeneity and three-dimensional structure.

Excess Solar Absorption in Cloudy Skies

Tropical Western Pacific Warm Pool

C95, R95 and PV95 employed independent sets of satellite, aircraft and surface observations for the tropical Pacific oceans, particularly over the warm pool in the western Pacific. They adopted the cloud forcing approach, in which the radiation data are separated into fluxes for clear skies and cloud forcing (difference between the average cloudy sky flux and the clear sky flux).

The picture that emerges from the three studies for the tropical western Pacific Ocean is as follows. The annual average clear skies absorb about 20% of the solar radiation (40). The annual average cloudy skies (clear plus overcast skies) absorb about 28%. The net result is that clouds enhance atmospheric absorption by about 8%. These findings are shown in Figure 5 in terms

of annual mean cloud forcing of the surface, the atmosphere and the surface-atmosphere column (TOA). The TOA cloud forcing, $C_s(\text{TOA})$, of -65 W m^{-2} is from the Earth Radiation Budget Experiment (ERBE) measurements; the surface cloud forcing, $C_s(\text{S})$, of -100 W m^{-2} is from R95. Thus, according to these studies, on average, clouds above the warm pool (compared to clear skies) enhance the reflection to space of solar radiation by 65 W m^{-2} , enhance the atmospheric absorption by 35 W m^{-2} , and reduce the solar radiation absorbed at the surface by 100 W m^{-2} .

The excess absorption is also indicated by another index, the cloud forcing ratio, f , where:

$$f = \frac{C_s(\text{S})}{C_s(\text{TOA})} = \frac{C_s(\text{S})}{C_s(\text{S}) + C_s(\text{Atm})} \tag{Eq. 6}$$

where $C_s(\text{Atm})$ is the atmospheric cloud forcing, i.e., effect of clouds on atmospheric solar absorption. If $C_s(\text{atm}) > 0$, then clouds enhance absorption over that in clear skies. It should also be noted that $C_s(\text{TOA})$, obtained by satellite radiation budget observations, is simply the sum of the surface and atmospheric cloud forcing. Thus, it follows that $f > 1$ indicates $C_s(\text{Atm}) > 0$. The GCMs used in the C95 study estimate $f \approx 1.0$. Hence, values of $f > 1$ in data would indicate excess absorption. The C95, R95 and PV95 studies independently obtain $f = 1.5$ for the warm pool. Other studies that have addressed the same problem have not come up with any major inconsistencies in the C95, R95 and PV95 studies, at least as far as the warm pool excess absorption is concerned.

– Waliser et al. (41) used open ocean data for the warm pool and showed that $C_s(\text{S}) \approx 100 \text{ W m}^{-2}$, almost identical to the R95 result. Likewise, Chou and Zhou (42) obtain a similar surface cloud forcing for a tropical western Pacific island station.

– Li et al. (27) employed satellite and surface data and obtain a tropical cloud forcing ratio in excess of 1.5. However, these authors conclude that poor data sampling in the tropics preclude definitive conclusions.

Since inferences about the excess absorption (at least in the C95 and R95 study) relies on satellite radiation budget measurements,

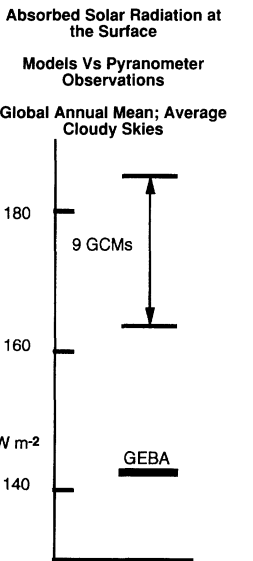


Figure 4. Comparison of the global annual mean shortwave absorbed at the surface as estimated from models and observations. The arrow indicates the range of values computed by nine general circulating models (GCMs). Observations are obtained from the Global Energy Balance Archive (GEBA). Adopted from results presented in Wild et al. (10).

a systematic uncertainty in the ERBE TOA cloud forcing would contribute to a spurious excess absorption. This is unlikely due to the following reasons:

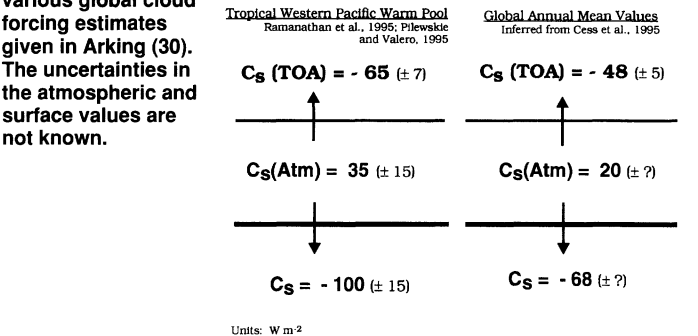
– ERBE had three independent satellites with different orbital characteristics; TOA annual mean cloud forcing from these three satellites agree to within 7 W m^{-2} (31 and R95). While individual pixel scale cloud forcing may have large uncertainties, diurnal-annual mean values are more accurate.

– The PV95 study does not use any satellite data, but simply relies on aircraft measurements above and below the clouds and arrives at the same f value, and excess absorption. However, the role of three dimensional cloud effects in the aircraft data has been raised as an uncertainty in the aircraft data (24; see response by 26).

At any rate, any residual uncertainty will soon be resolved with the launch of another but more advanced radiation budget instrument (the NASA-CERES instrument) in 1997.

Global Atmospheric Solar Absorption
The C95 study also analyzed data for six land stations both in the Northern

Figure 5. Excess absorption in terms of cloud forcing: Annual-diurnal mean values are presented. C_s is the shortwave cloud forcing, which is the difference between the average cloudy sky flux and the clear sky flux. $C_s(\text{TOA})$, $C_s(\text{Atm})$ and $C_s(\text{S})$ are, respectively, the TOA cloud forcing which represents the cloud forcing for the surface plus atmospheric column, the atmospheric cloud forcing and the surface cloud forcing. The warm pool uncertainties are inferred from R95 and denote 2σ values. For the global values, the TOA cloud forcing is from Harrison et al. (29), and the uncertainty in TOA cloud forcing is obtained by comparing the various global cloud forcing estimates given in Arking (30). The uncertainties in the atmospheric and surface values are not known.



Hemisphere and the Southern Hemisphere and found a cloud forcing ratio of about 1.5. Based on this, C95 suggested that the excess absorption is a global phenomenon. If the C95 results are valid globally, it has the following implications:

The global $C_s(\text{TOA})$ from ERBE is about 50 W m^{-2} . Hence, $f = 1.5$ implies that the surface cloud forcing, $C_s(\text{S})$, is 75 W m^{-2} yielding an excess absorption, i.e., $C_s(\text{Atm})$, of 25 W m^{-2} . This is about 8% of the TOA solar insolation; i.e., cloudy skies enhance atmospheric solar absorption from 20 to 28%, consistent with the atmospheric solar absorption deduced from the global GEBA data set by Ohmura and Gilgen (6) in Figure 3.

Discrepancies in Clear Sky Solar Absorption

Arking (11) used the GEBA data sets, in conjunction with TOA fluxes, and deduced the atmospheric solar absorption. Consistent with the recent studies, Arking showed that the observed all sky absorption is greater than the model estimates by about 25 to 30 W m^{-2} . However, his analyses suggest that this excess absorption was not related to the clouds, but more related to the atmospheric water vapor content. The one major difference between C95 and Arking’s study is that Arking uses monthly mean data whereas C95 uses hourly mean data. On monthly averages, water vapor, clouds, aerosols could be correlated, but the extent of the correlation as well as the sign may vary from region to region. Thus it is difficult to attribute cause and effect from monthly mean data. Further considerations to note are:

– For the tropical Pacific, calibrated radiometric data collected from a research vessel and two aircraft (one flying in the boundary layer and another at the tropopause) were used to estimate the atmospheric and surface absorption for clear skies (40, 43). The observational estimates from these data, sampled at 1 Hertz, varied between 19.5 to 20.5%, while the model values (that employed simultaneous observations of vertical soundings of water vapor and ozone) ranged between 19 and 21%, indicating excellent agreement for clear skies.

– C95 found that the observed clear sky surface absorption for Boulder, Wisconsin and American Samoa agreed to within a few percent of model values.

– However, Charlock and Alberta (44) used satellite data and surface observations from the Department of Energy’s Atmospheric Radiation Measurement site in Oklahoma to determine that clear sky calculations of atmospheric absorption significantly underestimate observations, and that this discrepancy is the same in cloudy skies.

SUGGESTIONS FOR RESOLVING THE DISCREPANCY

A missing heat source of the order of 25 W m^{-2} has significant implications for global climate, as indeed suggested by a GCM

study (45). It is imperative to identify the source for the discrepancy between the models and the observations. There are two avenues for further research:

- Undertake a detailed analysis of why some aircraft data (22, 23, 39) do not find the excess absorption in cloudy skies, while others observe a significant amount (20, 38, 46 and PV95). We focus on the aircraft studies, primarily because they offer a more direct method (than the satellite derived estimates) for observing absorption within the atmosphere. For this reason, we do not consider coincident satellite and surface measurements (11, 27, 44, C95) that also arrive at conflicting conclusions.

- Identify the source, spectral nature, and physics of the excess absorption.

A Detailed Study of the Aircraft Data

It is important to go back to the rich source of available aircraft studies to understand the source of the discrepancy. Such an attempt is beyond the scope of this paper, but we point to some promising avenues of research.

One of the first definitive aircraft studies suggesting a large excess absorption in clouds was by Twomey and Cocks (38). They employed a four-channel spectral radiometer at wavelengths: 0.75 μm , 1. μm , 1.2 μm and 2.25 μm . Their data revealed no excess absorption at 0.75 μm and significant excess absorption in the three near-IR regions. For example, the inferred absorption coefficients at the three near-IR wavelengths were factors of three to ten larger than the bulk optical properties of liquid water. It is important to note that this study did not find excess absorption in the 0.75 μm region. Note that if the inferred absorption is due to artifacts such as radiation leaking through the sides of clouds, then the inferred excess absorption should have been comparably large in the 0.75 μm region.

Another important study is the stacked aircraft study by Ackerman and Cox (22) during the Summer Monsoon Experiment (SMONEX) over the Arabian Sea. This was perhaps the first study to attempt the removal of the so-called three-dimensional radiative transfer effects in the observed flux differences. The justification is:

- With the stacked two aircraft, the net (up minus down) irradiances at the two flight altitudes (above and below the cloud) are differenced to obtain the energy converging (i.e., absorption) into the cloud. However, since some of the radiation escapes

from the sides of clouds, the net flux difference at the two flight altitudes includes two components: a) absorption within clouds; and, b) radiation leaking through the sides.

In order to remove the side radiation component, Ackerman and Cox (22), adopted a normalization procedure derived from the Monte Carlo model results of Welch et al. (47) for hypothetical three-dimensional clouds. The fundamental assumptions of this normalization procedure are as follows:

- There should be no in-cloud absorption in the 0.3 to 0.7 μm . This is based on the fact that pure water has no absorption features in this spectrum. By definition, it asserts that any flux difference in this region is due to radiation leaking through the sides and should be removed from the data.

- Furthermore, three-dimensional effects in the 0.3 to 0.7 μm regions are assumed to be identical to that in the 0.7 to 4 μm region. As a result, the percent absorption that is removed from the 0.3 to 0.7 μm region should be removed from the 0.7 to 4 μm region also.

We will first report Ackerman and Cox's observations of the absorption without the normalization procedure. The case is a stratocumulus field over the Arabian Sea and the absorption is between the flight altitudes of 347 and 5175 m. The broad band region considered is 0.3 to 2.8 μm . The flux convergence was obtained by averaging the data over a 24 km length scale.

Cloud absorption value without the normalization adjustment for the 0.3 to 2.8 μm region is 16%. The clear sky absorption in the broad band region of 0.3 to 2.8 μm for the same layer of 347 to 5175 m is about 8%. Comparing this to the 16% absorption for the cloud layer, the actual data show an absorption enhancement of 8% in the cloudy layer, when compared with clear sky values. However, after the normalization procedure, the adjusted cloud absorption in the 0.3 to 2.8 μm region is 9%, almost identical to that in the clear skies. The following are important points:

- The atmosphere is not strictly nonabsorbing in the 0.3 to 0.7 μm region, e.g., see computed spectral absorption in clear skies (Fig. 6). These absorption features can be enhanced due to multiple scattering both within clouds, and by cloud to cloud scattering.

- The excess absorption without the adjustment is very close to the findings of the studies of the C95, R95 and PV95.

- The stacked aircraft study of PV95 does not apply the edge effect correction; in other words, the significant excess absorption values reported by the PV95 stacked aircraft data for the tropical Pacific clouds (thick convective cirrus anvils) are consistent with the stacked aircraft data (but unadjusted) for the tropical Arabian Sea clouds reported by Ackerman and Cox. This is an important point since PV95 employ a thermistor bolometer radiometer which has very different instrumental characteristics from the Eppley pyranometers employed by Ackerman and Cox. The agreement of the unadjusted data suggests that the controversy is not necessarily due to instrumental errors, but can be narrowed to the data reduction procedure.

- According to Ackerman and Cox (Fig. 1 in 22), the edge effects are important only for about 2.5 km from the side, and negligible for the rest of the interior of the clouds. Pilewskie and Valero (26) report that the PV95 data observed clouds that were many hundred kilometers in diameter (or width) such that the edge effects (employing Fig. 1 in 22) should be minimal.

- For a single isolated cloud of finite width, there is no doubt that the edge effects are important and the data should be adjusted. But, if each flight leg of the irradiance measurements included tens of such clouds and the data were averaged over tens of individual finite size clouds, we would expect that the flight level measurements would capture the photons leaking through the sides. This problem needs to be studied carefully with models of realistically shaped clouds and their fields.

After the Ackerman and Cox study, other studies began ap-

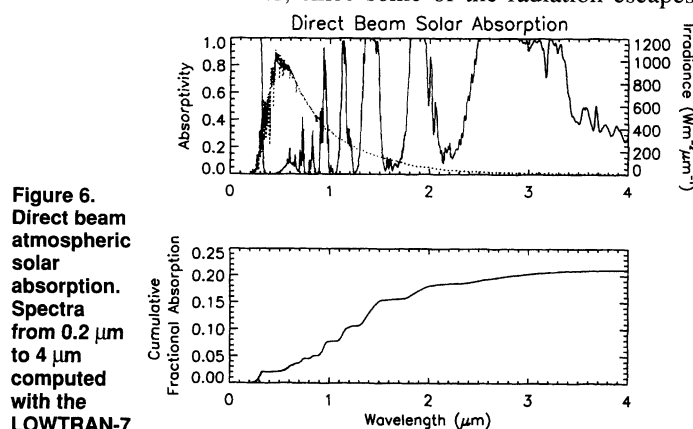


Figure 6. Direct beam atmospheric solar absorption. Spectra from 0.2 μm to 4 μm computed with the LOWTRAN-7 model using a middle latitude summer profile and a solar zenith angle of 60°. The top panel gives the absorptivity (solid line, left axis), and the cosine-solar-zenith-angle weighted solar irradiance at the top of the atmosphere (dotted line, right axis). The bottom panel gives the cumulative atmospheric absorption as a fraction of the total top-of-atmosphere solar irradiance. The gases that are responsible for the most prominent absorption features areas follow. Water vapor dominates the absorption for wavelengths greater than 0.8 μm with the exception of the significant contribution by CO₂ at 2.7 μm and its lesser contributions at 1.4, 1.6, and 2.0 μm . Ozone absorbs strongly in the ultraviolet region from about 0.2 to 0.3 μm with weaker absorption at visible wavelengths, in particular those less than about 0.68 μm . Oxygen has bands centered at 0.76, 0.69 and 0.63 μm , and is responsible for the strong absorption in the far ultraviolet, particularly wavelengths less than about 0.2 μm .

plying their normalization procedure (e.g., 23). It seems that, without the Ackerman and Cox normalization procedure, most of the aircraft data (at least those that employ calibrated instruments such as those in 22, 20, 37 or PV95) would be in agreement.

For example, Herman (37), Herman and Curry (20) and Kondrati'ev et al. (21), who do not apply the normalization procedure to their aircraft data, suggest significant excess absorption. For the Arctic stratus, pyranometer measurements obtained by Herman (37) reveals 5% absorption in the visible and 9% in the near-IR. Herman's studies (37) were repeated in June 1980 for the Arctic stratus by Herman and Curry (20) with a silicon detector radiometer, and they report that the computed average (over several flights) cloud-layer absorption was about 10.4% while models yielded (for the same cloud thickness and type) an absorption of only 3.6%. Kondrati'ev et al. (21) employ several years of aircraft data to conclude: "These data demonstrate the existence of substantial SWR absorption by clouds in a visible region of the spectrum" (translated from Russian).

We conclude with an important study by King et al. (39) which, instead of measuring the flux differences, attempted to deduce the spectral absorption coefficients directly from the data. They developed a scanning spectral radiometer for aircraft and used the angular distribution of the radiation within a cloud to infer the single scattering albedo of marine stratus clouds off the coast of California. The data reduction procedure requires the assumption that "the cloud is non-absorbing at some wavelengths in the visible region." Their data reveal that "in the wavelength region between 1.6 and 2.2 μm , our measurements of excess absorption are consistent with the observations of Twomey and Cocks (38) . . ." Mel'nikova and Mikhaylov (46) adopt the King et al. (39) approach, but re-derive the data reduction procedure by relaxing the assumption that the cloud be nonabsorbing in the visible region. The refined algorithm, when applied to spectral data, reveals significant absorption in both visible and near IR regions.

In summary, a consistent pattern emerges from all of the aircraft studies. The data reduction procedure generally assumes that clouds have no absorption in the visible region. When this assumption is relaxed, most of the data show large values of excess absorption across the entire solar spectrum. This is an encouraging picture, since it reduces the controversy to an examination of the normalization procedure, instead of a detailed re-examination of all of the aircraft data and the instrumentation used.

The Source, the Spectral Nature and the Physics of Excess Absorption

Clear Skies: We will consider Arking's (11) suggestion that the source of the missing 25 to 30 W m^{-2} is related to water vapor. First, the analyses should be repeated with hourly or daily data, instead of the monthly means employed by Arking (11). In the mean time, let us explore Arking's (11) conclusion that water vapor is the source. This can arise from one of the two following possibilities: a) uncertainties or missing physics (continuum) in water vapor absorption; or b) missing physics in aerosol absorption, since aerosol concentrations (the aqueous ones) can increase with the relative humidity.

Water Vapor. Figure 6 shows the spectral nature of the modeled clear sky absorption. Water vapor absorption accounts for about 70% of the total absorption. Out of the total of 60 to 65 W m^{-2} global-diurnal mean clear sky absorption computed by models (34), water vapor contributes about 40 to 45 W m^{-2} . Thus, for water vapor to explain about 25 W m^{-2} more, the computed absorption has to be systematically smaller by about 50 to 60%. We are not aware of any reported study which suggests such large errors of the order of 50%. In summary, systematic uncertainties in vapor absorption might contribute to excess ab-

sorption, but they are unlikely to be a major contributor.

Aerosols. With respect to aerosols, it is nearly impossible to explain a global effect of 25 W m^{-2} from uncertainties in aerosol absorption. However, there has been a long history of anomalous effects linking aerosols with large unexplainable absorption (19, 20, 36, 37), at least for continental regions. Even if the global average of such anomalous effects may be small, it is fundamental to understand them since, if real, the effects can be amplified within clouds.

Robinson's (19) surface observations, one of the more extensive and well documented early studies, revealed anomalous absorption ranging between 2 to 10% (of the TOA insolation.), which he attributed to aerosols (for average TOA insolation of 342 W m^{-2} , this range translates to 7 to 34 W m^{-2}). We paraphrase his conclusion, which applies equally to many of the papers cited on the topic of excess absorption:

"We conclude, in spite of our ability to explain certain features of the records, that measurements of solar radiation at the ground in cloudless conditions indicate the frequent and geographically widespread occurrence of atmospheric absorption additional to that attributable to the gases H_2O and CO_2 . This absorption is correlated with the scattering associated with atmospheric aerosol."

Similar findings related to anomalous aerosol absorption can be found in Roach (36). In summary, aerosols in regions far away from urban areas may contribute to excess absorption on the order of a few percent.

Cloudy Skies: The answer may very well be some combination of: truly anomalous behavior in clouds and aerosols; cloud impurities (i.e., deviation from pure water properties); absorption by large drops (48, 49); cloud morphology (50, 51); and enhancement of the photon mean free path by three-dimensional effects (52, 53). All of these effects, which would enhance absorption in clouds, are largely ignored in radiation models used in climate studies, at least until recently. The radiation models have a long way to go before they can resolve the issues raised here. The future looks promising, particularly since our ability to make observations of the cloud-atmosphere-radiation interactions are in step with our capability to realistically model these phenomena.

CONCLUSIONS

Arrhenius' paper continues to be an excellent reference work for learning the approach of using direct observations for modeling. Step by step, Arrhenius demonstrates how to build a radiation and an energy balance model directly from observations. Arrhenius' techniques for interpreting Langley's radiometric data for this purpose is a lesson worth learning even now. The successes of Arrhenius' model are many, even when judged by modern day data and computer simulations.

Arrhenius introduced the fixed relative humidity assumption to model the water vapor feedback, thus anticipating the pioneering radiative-convective model study of Manabe and Wetherald (14). We demonstrate that the water vapor feedback amplified the surface warming in Arrhenius' model by about 30%. This is probably an accurate amplification factor, given the fact that his model is a hybrid between the radiative equilibrium model, the fixed-dynamical heating rate model and the energy balance model. Second, we show that, his model overestimated the CO_2 radiative effect by about a factor of 2.5, primarily because the model severely overestimates the opacity of the CO_2 bands in the 6 to 8 μm region. Water vapor is the stronger absorber in this region and it is extremely difficult (if not impossible) to separate the H_2O effect from the CO_2 absorption solely from surface observations. Arrhenius could not have attacked this issue any better, particularly since quantum mechanics was not yet discovered.

The second part of the paper focuses on the atmospheric solar absorption which was considered in great detail by Arrhenius. Global models suggest that the average cloudy atmosphere absorbs about 20% of the solar radiation at the top of the atmosphere, while at least six recent studies have concluded that it should be about 28%. The large magnitude, the lack of an adequate physical explanation for the excess absorption and conflicting results from various studies have become the source of a controversy. We focus our attention on the aircraft studies, since the aircraft platform offers the most direct way of observing cloud solar absorption. We narrow the controversy to the normalization procedure for reducing aircraft data by showing that without the 1980s normalization procedure, most if not all of the aircraft data will consistently show large excess absorption in cloudy skies.

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We do not recommend a premature abandonment of the normalization procedure, for it requires the acceptance of the excess absorption to be in the visible region (0.4 to 0.7 μm) as well. If there is indeed significant excess absorption in the 0.4 to 0.7 μm region, the excess absorption is a truly anomalous phenomenon, for no known physics would yield the required absorption in the visible. Steps required to resolve the controversy include the following: i) Conduct detailed experimental and modeling studies to establish the validity of the normalization procedure developed in the 1980s; and ii) Undertake laboratory and in situ observations to establish the source, if any, of absorption in the 0.4 to 0.7 μm spectral region in cloudy atmospheres, beyond the trace amount of absorption due to ozone, oxygen and water vapor.

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